

v0 Dry-Run Dataset Report

Eight-Regime Surrogate Dataset Generation Test

Codex-assisted surrogate data review

June 18, 2026

Abstract

This report reviews the v0 dry-run dataset generation test for the American option risk-surface project. The dry run generated a small eight-regime package using the validated baseline American Crank–Nicolson / projected SOR solver. It produced one compressed `.npz` file, audit manifests, diagnostics, and optional review figures. It did not generate the full 288-regime dataset, train a surrogate model, create `.pt` files, or approve production Greek labels. The dry-run review decision is `READY_FOR_V1_SMALL_GRID_PLANNING`.

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1 Purpose of v0 Dry Run

The v0 dry run is the first controlled execution after the surrogate dataset construction plan. Its purpose is to test the proposed pipeline on eight regimes before any larger generation step. The dry run checks whether solver outputs, continuation premium, boundary metadata, Greek masks, LCP diagnostics, split metadata, and file manifests can be assembled into one auditable package.

This is a dry-run generation test only. It is not the final dataset, not a neural-surrogate training stage, and not approval for broad production-scale sweeps.

2 Roadmap and Construction-Plan Link

The compressed roadmap places this work in Stage 1: v0 dry-run dataset plus dry-run review. The preceding dataset construction plan named the full 288-regime grid, split policy, failure rules, and eight-regime dry-run subset. This report records the outcome of executing that subset.

The solver source remains the validated classical benchmark:

```
solver_name=american_crank_nicolson_psor_price
solver_variant=baseline_cn_psor
```

The numerical method and American-option setting follow the project validation ladder and the classical option-pricing references already integrated into the solver reports (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

3 Exact Eight Regimes

Dry ID	Regime ID	Type	T	σ	r	q	Split
dry_01	put_T100_s020_r005_q003	put	1.0	0.20	0.05	0.03	train
dry_02	put_T100_s060_r005_q003	put	1.0	0.60	0.05	0.03	train
dry_03	put_T100_s020_r001_q003	put	1.0	0.20	0.01	0.03	train
dry_04	put_T200_s040_r010_q010	put	2.0	0.40	0.10	0.10	test
dry_05	call_T100_s020_r005_q006	call	1.0	0.20	0.05	0.06	train
dry_06	call_T100_s020_r005_q010	call	1.0	0.20	0.05	0.10	train
dry_07	call_T100_s060_r005_q006	call	1.0	0.60	0.05	0.06	train
dry_08	call_T100_s020_r005_q000	call	1.0	0.20	0.05	0.00	test

All regimes used $K = 1$, $S_{\max} = 4$, and $M = N = 120$. The no-dividend call row is a control regime, not dividend-call early-exercise evidence.

4 Feature, Label, and Mask Schema

The stored feature matrix uses:

$$x = (\log(S/K), \tau/T, r, q, \sigma, T, \text{is_call}).$$

The package stores reliable price-style labels and diagnostic labels separately:

- core dry-run labels: `value_over_K`, `payoff_over_K`, and `premium_over_K`;

- diagnostic labels: `exercise_indicator` and `boundary_spot_over_K`;
- cautioned Greek diagnostics: `delta` and `scaled_gamma`.

Masks are included for payoff-kink-near, boundary-near, maturity-row, strict-interior, Gamma-allowed, Delta-allowed, exercise-region, and continuation-region nodes. These masks are mandatory metadata. They are especially important because Gamma remains fragile near kinks, maturity rows, and extracted boundary bands.

5 Diagnostics and Acceptance Checks

A regime is accepted into the dry-run arrays only when PSOR converges, obstacle violation is at most 10^{-8} , equation violation is at most 10^{-6} , complementarity product is at most 10^{-6} , and required metadata and masks are present. Failed regimes would remain visible in the CSV manifests with `review_required`; they would not be silently included as accepted label rows.

Metric	Dry-run result	Interpretation
Regimes solved	8	Exactly the approved dry-run subset.
Accepted regimes	8	All regimes passed the dry-run acceptance checks.
Accepted sampled rows	41,624	5,203 rows per accepted regime.
Train rows	31,218	Six train regimes.
Test rows	10,406	Two test/control regimes.
Maximum obstacle violation	0.0	Payoff obstacle respected in accepted regimes.
Maximum equation violation	4.90×10^{-8}	Below the 10^{-6} diagnostic threshold.
Maximum complementarity product	5.34×10^{-10}	Below the 10^{-6} diagnostic threshold.
Maximum PSOR iterations	41	Highest effort occurs in a high-volatility put regime.
Total solver runtime	about 0.82 seconds	Small dry run is computationally manageable.

6 Output Files and Manifests

Output	Type	Purpose
<code>results/04_surrogate_dataset/v0_dry_run/dataset_v0_dry_run.npz</code>	<code>.npz</code>	Accepted-row dry-run arrays only.
<code>results/04_surrogate_dataset/v0_dry_run/regime_manifest.csv</code>	CSV	Regime parameters, convergence, and acceptance status.

Output	Type	Purpose
results/04_surrogate_data et/v0_dry_run/diagnostic_s ummary.csv	CSV	PSOR, LCP, boundary, Greek, and runtime diagnostics.
results/04_surrogate_data et/v0_dry_run/split_assign ment.csv	CSV	Regime-level split assignment and row counts.
results/04_surrogate_data et/v0_dry_run/schema_snaps hot.csv	CSV	Feature, label, mask, and audit-field schema.
results/04_surrogate_data et/v0_dry_run/output_manif est.csv	CSV	Output audit manifest for all dry-run artifacts.

The compressed package contains the keys `X`, `y_value`, `y_payoff`, `y_premium`, `y_exercise_indicator`, `y_boundary`, `y_delta`, `y_scaled_gamma`, `masks`, `regime_index`, `feature_names`, `label_names`, `mask_names`, `audit_numeric`, `audit_numeric_names`, `regime_ids`, `dry_run_ids`, and `split_names`.

7 Figure Interpretation

Three optional figures were created for human review. They are diagnostic figures only, not stress-map datasets.

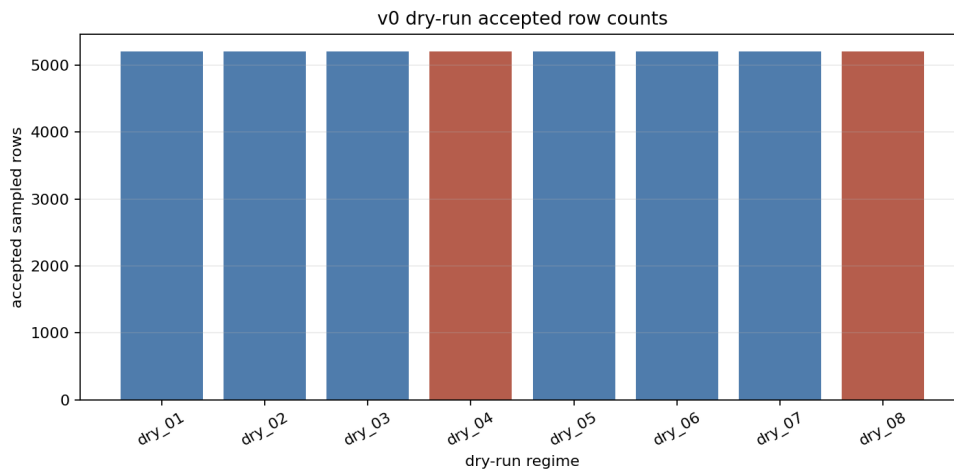


Figure 1: Accepted sampled rows by dry-run regime. Each accepted regime contributes 5,203 sampled rows in the reporting region.

8 Dry-Run Review Decision

The dry-run review decision is:

READY_FOR_V1_SMALL_GRID_PLANNING

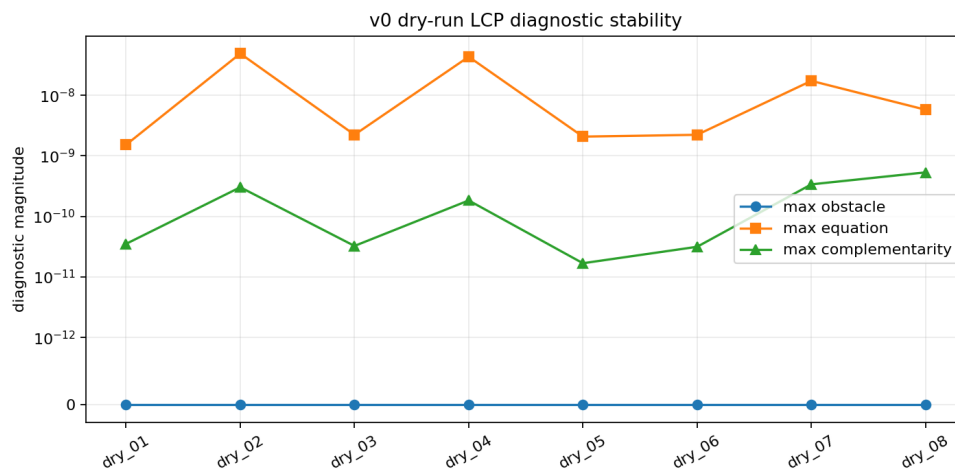


Figure 2: Obstacle, equation, and complementarity diagnostics by regime. The equation residuals remain below the dry-run threshold.

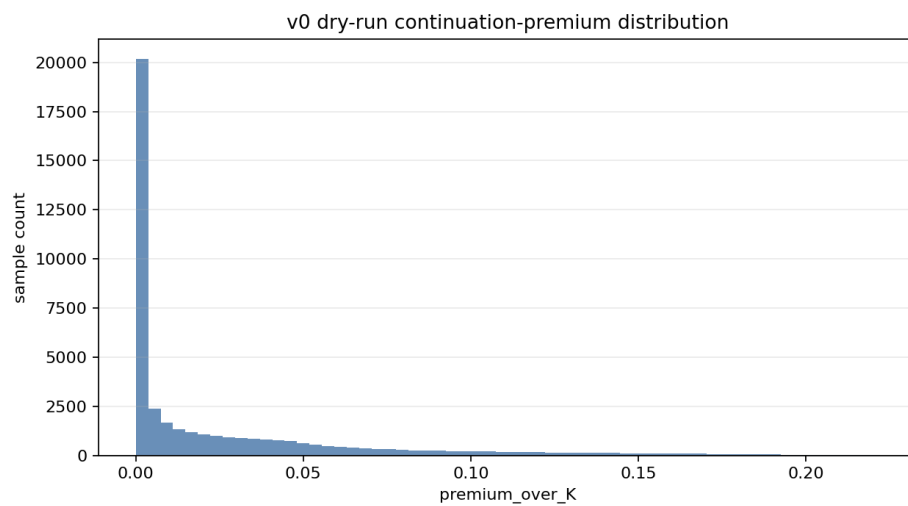


Figure 3: Distribution of continuation premium over accepted rows. This checks the dry-run label pipeline without creating broad surfaces.

This means the eight-regime pipeline is ready for a v1 small-grid planning step. It does not mean the full dataset is approved for generation, and it does not approve surrogate training.

9 Limitations

- The package covers eight regimes only, not the planned 288-regime grid.
- Boundary labels remain threshold-based diagnostics, not exact analytical free boundaries.
- Delta and scaled Gamma are diagnostic fields. Gamma remains high caution and should be used only with masks.
- Runtime observations from eight regimes do not prove large-scale generation cost.
- The no-dividend call control has no boundary found, which is expected control behavior rather than a failed dividend-call boundary.

10 What This Dry Run Supports

The dry run supports the claim that the approved eight-regime generation pipeline can assemble features, reliable price and premium labels, diagnostic labels, masks, solver diagnostics, split metadata, and manifests into one auditable package. It also supports moving to a v1 small-grid planning discussion.

11 What This Dry Run Does Not Support

The dry run does not support final dataset claims, neural surrogate training, production Greek labels, broad parameter-sweep conclusions, or final paper claims about all regimes. It is a pipeline test and review gate.

12 Recommended Next Step

The recommended next step is a v1 small-grid planning gate. That gate should decide whether to generate all 288 planned regimes, whether any additional higher-grid confirmation is required, and whether Delta or Gamma should remain metadata-only in the first generated version.

References

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